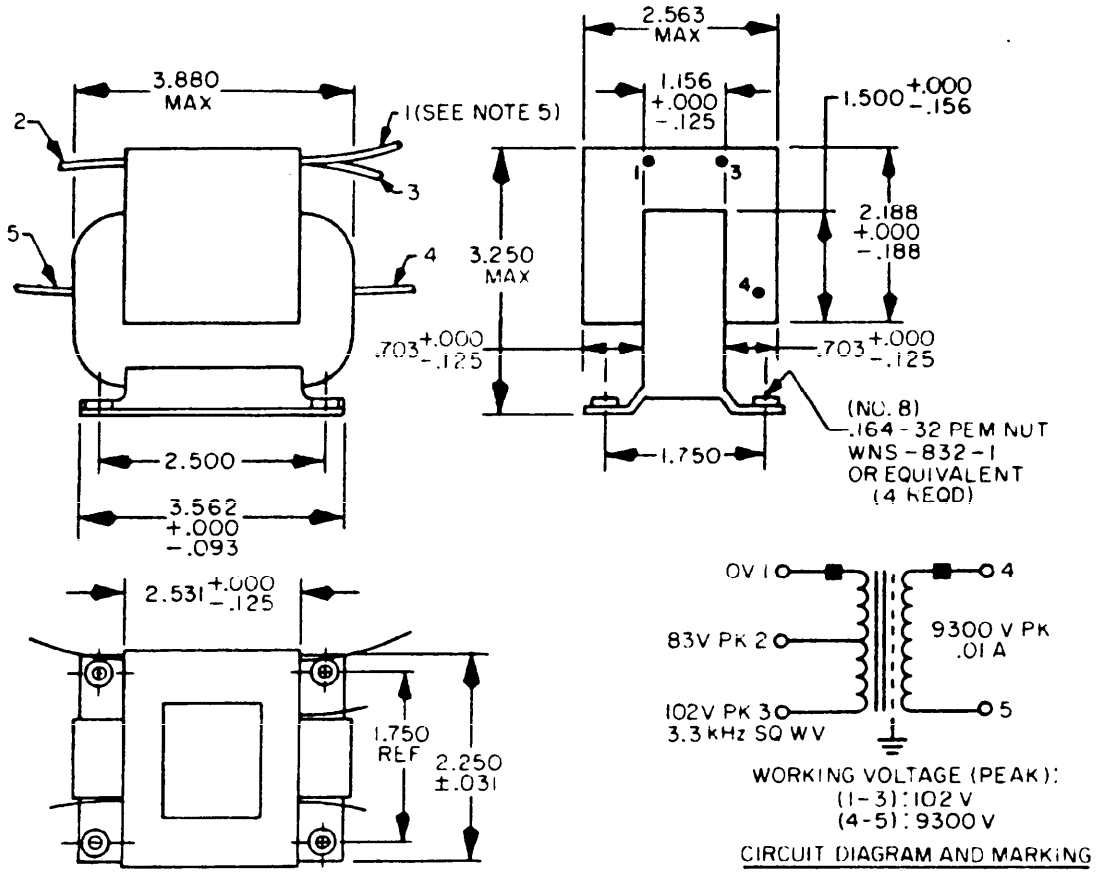
MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, 112 VA, STEP-UP, TF5S037Z

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of Specification MIL-T-27.



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.031	.79	.188	4.78	1.750	44.45	2.531	64.29
.093	2.36	.703	17.86	2.188	55.58	2.563	65.10
.125	3.18	1.156	29.36	2.250	57.15	3.250	82.55
.156	3.96	1.500	38.10	2.500	63.50	3.562	90.47
.164	4.17					3.880	98.55

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Tolerance on mounting position dimensions is ±.016 (.41 mm).
4. Marking shall be on top of the case and sides if required.
5. All wire leads shall be a minimum of 8.0 inches (203.2 mm) in length.
6. Mounting surface and threads shall be free of encapsulation material.

FIGURE 1. Dimensions and configuration.

FSC 5950

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REQUIREMENTS: (When numbers in parentheses, i.e. (1-2), are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:
- Primary voltage, 3.3 kHz square wave ±20%.
  - (1-2): 83 volts, peak.
  - (1-3): 102 volts, peak.
- Primary current:
  - (1-2): 1.31 amperes.
  - (1-3): 1.10 amperes.
- VA: 112
- Secondary voltage (4-5): 9300 volts, peak.
- Secondary current (4-5): 0.01 ampere.
- Working voltage:
  - (1-3): 102 volts, peak.
  - (4-5): 9300 volts, peak.
- Operating temperature range: -55°C to +130°C.

Design and construction:
- Dimensions and configuration: See figure 1.
- Duty cycle: Continuous.
- Case: Encapsulated.
  - Material: Epoxy
- Terminals: Insulated wire leads per MIL-W-81044/6
  - Terminal length: 8.0 inches minimum.
  - Terminal diameter:
    - (1, 2 and 3): 0.041 inch $\binom{+.000}{-.004}$
    - (4 and 5): 0.026 inch $\binom{+.000}{-.004}$
- Weight: 2.5 pounds maximum.

Thermal shock: Method 107 of MIL-STD-202, test condition A-1.

Terminal strength: Method 211 of MIL-STD-202, test condition A, 2.0 pounds.

Shock: Method 213 of MIL-STD-202, test condition H.

Dielectric withstanding voltage (at sea level):
- (1-GND): 600 volts, rms.
- (1-4): 18 kilovolts, rms.
- (4-GND): 14 kilovolts, rms.

Electrical characteristics:
- No load: With 102 volts, peak and 3.3 kHz square wave across (1-3):
  - Current in (1-3): 0.300 ampere maximum.
  - Power in (1-3): 30.6 watts maximum.
  - Voltage across (4-5): 9440 volts, peak.
- Rated load: With 102 volts, peak and 3.3 kHz square wave across (1-3):
  - Power in (1-3): 119 watts maximum.
  - Current in (1-3): 1.16 amperes maximum.
  - Voltage across (4-5): 9300 volts, peak ±5%.
  - Current in (4-5): 0.01 ampere.
- Electrostatic shielding: Voltage ratio: 10 to 0.88 at 3.3 kHz square wave (15 dB).

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DC resistance:
(1-3): 0.277 to 0.373 ohm.
(4-5): 2.813 to 3.807 kilohms.
Polarity: Additive, with terminals 3 and 4 connected.
Temperature rise: 50°C with 102 volts, peak, 3.3 kHz square wave across:
(1-3) at an ambient temperature of 80°C.

Marking location: See figure 1.

Part number: M27/139-01

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI, SG
Navy - EC, SH, OS
Air Force - 11, 17, 85, 99
DSA - ES

User activities:
Army - MU, WC, SG, ME
Navy - MC, AS, SH, OS
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5950-0536)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-T-27/139 TRANSFORMER, POWER 112 VA, STEP-UP, TF5S03ZZ

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

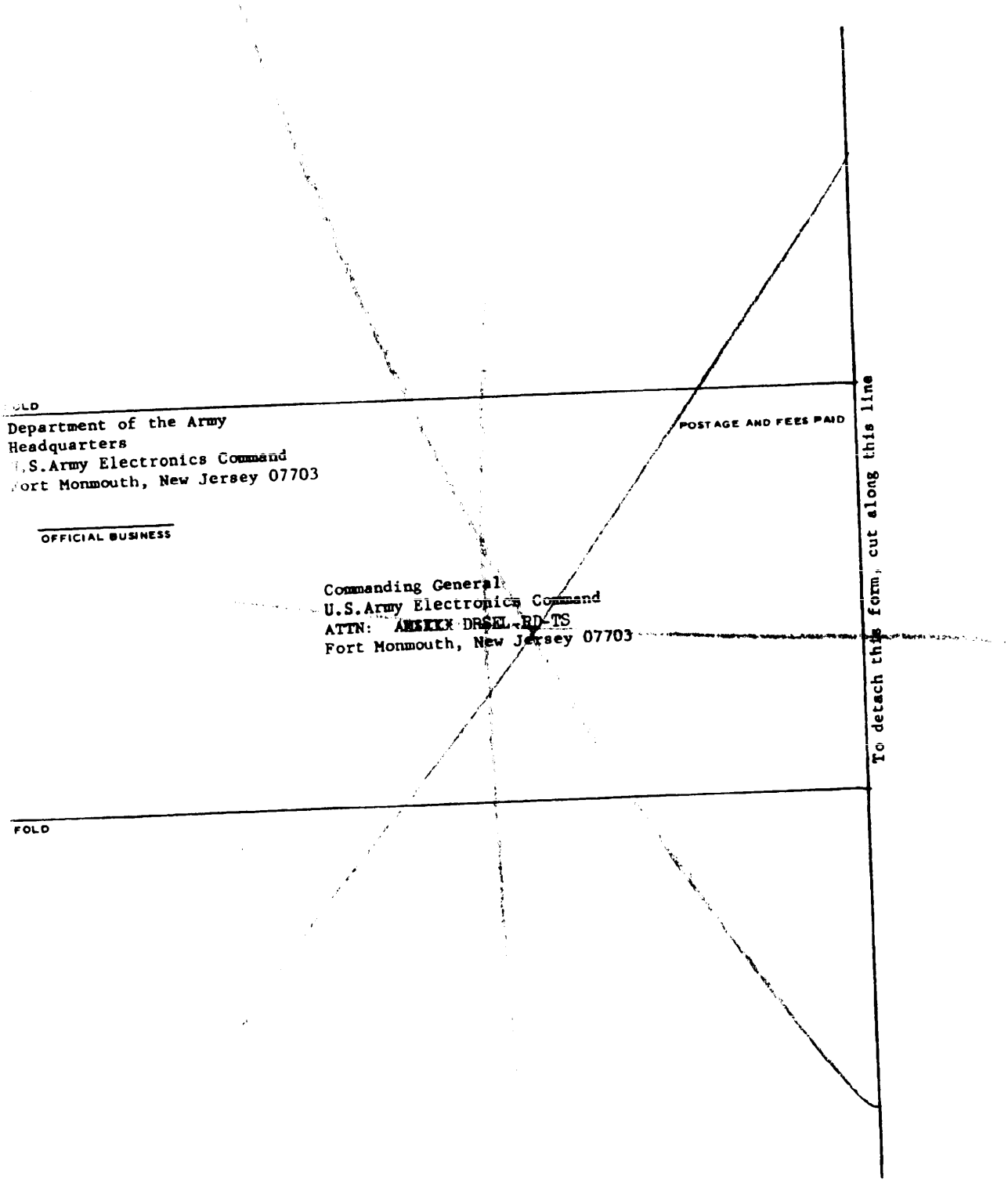
DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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